

Module 3 Introduction

Portfolio Performance Evaluation

Overview

This module builds on the earlier discussion of financial risk and return by introducing a systematic framework for evaluating portfolio performance. In practice, investors and portfolio managers are not interested only in whether a portfolio made a profit; they must determine whether the return achieved was appropriate given the level of risk taken, the time horizon involved, and the prevailing economic environment.

The module begins with the measurement of portfolio returns, examining how investment performance is calculated over single and multiple periods. It then introduces the concept of portfolio risk, distinguishing between total risk, systematic risk, and the role of diversification in reducing asset-specific uncertainty. Building on these foundations, the module explores risk-adjusted performance measures, such as the Sharpe and Treynor ratios, which allow investors to compare portfolios with different risk profiles.

The analysis then extends beyond nominal returns by considering real returns and the impact of inflation on purchasing power, highlighting why nominal gains may overstate true economic performance. Finally, the module introduces downside-risk and tail-risk measures, which provide a more realistic evaluation of portfolios when returns are asymmetric or subject to extreme losses.

By the end of the module, you will understand how professional investors evaluate portfolios using a multi-dimensional framework that integrates return measurement, risk assessment, and risk-adjusted performance analysis. This framework provides a practical toolkit for comparing investment strategies and assessing whether a portfolio delivers risk-appropriate and economically meaningful returns.

What will you be learning this week?

Key Concepts	Content
3.1 Measuring Portfolio Returns	Understand how investment performance is measured using Holding Period Return (HPR) and distinguish between arithmetic and geometric returns. Learn how

Key Concepts	Content
	compounding affects long-term performance and why the geometric mean provides a more accurate measure of multi-period wealth growth.
3.2 Understanding Portfolio Risk	Examine the concept of risk as return variability and distinguish between total risk, systematic risk (beta), and unsystematic risk. Understand how diversification reduces firm-specific risk but cannot eliminate market risk.
3.3 Risk-Adjusted Portfolio Performance	Evaluate how effectively portfolios convert risk into return using Sharpe and Treynor ratios. Learn when each measure is appropriate and how they differ in their treatment of total versus systematic risk.
3.4 Real Returns and Purchasing Power	Understand the difference between nominal and real investment returns using the Fisher relationship. Learn why inflation adjustments are necessary when assessing long-term investment performance and wealth preservation.
3.5 Downside Risk and Tail-Risk Measures	Analyse performance when return distributions are asymmetric or fat-tailed. Learn how Sortino Ratio, Conditional Value at Risk (CVaR), and Modified Sharpe Ratio focus on downside risk and extreme losses to provide a more

Key Concepts**Content**

	realistic assessment of portfolio risk.
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What will you be able to do by the end of this module?

This module develops your ability to evaluate investment performance using a structured analytical framework, linking portfolio returns to risk, inflation, and extreme market outcomes.

How Module 3 supports your Course Learning Outcomes

This module develops your ability to measure, interpret, and communicate financial risk using time series methods in R, supporting:

- **CLO 1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO 2:** Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.
- **CLO 4:** Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

How much time can you expect to spend on Module 3?

This workload is designed to be manageable and flexible (i.e. you don't need to complete everything in one sitting!).

Estimated learning hours

You will need to spend around 25 hours in total engaging with the learning materials, attending live sessions, studying, and working towards your assessment.